Homework 01
REAL VARIABLES I
Fall 2026

Due: 2026-08-30 19:00
Topics: Non-negative sums over arbitrary sets: limit theorems. Lebesgue outer measures, fat rationals, and countability

Instructions
Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

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1. (Monotone convergence for nonnegative sums)

IN CLASS

1.a (weight: 0.7) Let $\mathcal{J}$ be an arbitrary index set and let $(a_{j,k} \in \overline{\mathbb{R}}_+)_{j\in\mathcal{J},k\in\mathbb{N}}$ be an indexed set of non-negative extended reals.
Suppose that $a_{j,k} \uparrow a_j$ as $k \to \infty$, for every $j \in \mathcal{J}$. Show that $\sum_{j\in\mathcal{J}} a_{j,k} \uparrow \sum_{j\in\mathcal{J}} a_j$ as $k \to \infty$. Recall that $a_n \uparrow a$ as $n \to \infty$ means that $a_{n+1} \geq a_n$ for all n and $\lim_{n\to\infty} a_n = a$.

1.b (weight: 0.3) Provide an example of $(a_{j,k} \in \overline{\mathbb{R}}_+)_{(j,k)\in\mathbb{N}\times\mathbb{N}}$ such that, for every $j \in \mathbb{N}$, the limits $\lim_{k\to\infty} a_{j,k}$ and $\lim_{k\to\infty} \sum_{j\in\mathbb{N}} a_{j,k}$ exist, but

$$\lim_{k\to\infty} \sum_{j\in\mathbb{N}} a_{j,k} \neq \sum_{j\in\mathbb{N}} \lim_{k\to\infty} a_{j,k}.$$

2. (Dominated convergence for nonnegative sums)

IN CLASS

2.a (weight: 0.7) Let $(a_{j,k} \in \overline{\mathbb{R}}_+)_{(j,k)\in\mathbb{N}\times\mathbb{N}}$. Suppose that $\lim_k a_{j,k} = a_j$ for each $j \in \mathbb{N}$ and there exists a dominating profile $(g_j \in \overline{\mathbb{R}}_+)_{j\in\mathbb{N}}$ such that $\sum_{j\in\mathbb{N}} g_j < +\infty$, and $a_{j,k} \leq g_j$ for every $j \in \mathbb{N}$ and $k \in \mathbb{N}$. Show that

$$\lim_{k\to\infty} \sum_{j=1}^{\infty} a_{j,k} = \sum_{j=1}^{\infty} a_j.$$

2.b (weight: 0.3) Provide an example of $(a_{j,k} \in \overline{\mathbb{R}}_+)_{(j,k)\in\mathbb{N}\times\mathbb{N}}$, such that the limit $\lim_k a_{j,k}$ exists for each $j \in \mathbb{N}$, the limit $\lim_{k\to\infty} \sum_{j=1}^{\infty} a_{j,k}$ exists, but

$$\lim_{k\to\infty} \sum_{j=1}^{\infty} a_{j,k} \neq \sum_{j=1}^{\infty} a_j.$$

3. (Countable sets have null Lebesgue outer measure)

3.a (weight: 1.0)
Let $E \subset \mathbb{R}$. Define the Lebesgue outer measure $\mathcal{L}^*$ of $E \subset \mathbb{R}$ as follows:

$$\mathcal{L}^*(E) \coloneqq \inf\Big\{\sum_{j=0}^{\infty}(b_j - a_j) \;:\; \bigcup_{j=0}^{\infty}(a_j, b_j) \supseteq E\Big\}.$$

Suppose E is countable. Prove that $\mathcal{L}^*(E) = 0$.

> **Free hint.** Enumerate E and use a construction similar to the fat rationals.

4. (Countable and uncountable sets)

4.a (weight: 0.5) Let $\ell_c(\mathbb{N}; \mathbb{Q})$ be the set of all sequences $(a_n)_{n\in\mathbb{N}}$ with values in $\mathbb{Q}$ that have finite support; that is, $a_n = 0$ for all but finitely many $n \in \mathbb{N}$. Prove that $\ell_c(\mathbb{N}; \mathbb{Q})$ is countable.

> **Free hint.** First, prove countability of strings supported on $\{0, \dots, N\}$ for a fixed $N \in \mathbb{N}$.

4.b (weight: 0.5) Prove that $|\mathbb{R} \times \mathbb{R}| = |\mathbb{R}|$.